Raw Materials, Composition, Equipment, and Manufacturing Processes for Tractor Wet Brake Friction Plates

1. Introduction

Tractor wet brake friction plates operate in an oil-lubricated environment, requiring materials that provide stable friction coefficients, high wear resistance, and thermal stability under repeated engagement cycles. Unlike dry brake systems, wet friction materials must maintain performance while submerged in transmission/hydraulic oil, which serves the dual purpose of lubrication and cooling . The manufacturing of these plates involves sophisticated material formulation and specialized production processes to achieve the required performance characteristics.

2. Raw Materials and Composition

2.1 Paper-Based Friction Materials

The most common friction material for tractor wet brakes is paper-based friction material, which offers high dynamic friction coefficients and lower cost compared to sintered or carbon materials . The typical composition includes:

Fibrous Reinforcement (45-55% by weight):

- Aramid fibers
- Carbon fibers
- Composite reinforced fibers
- Cotton pulp (in some formulations)

Functional Raw Materials (approximately 20%):

- Friction modifiers

- Solid lubricants (graphite)

Filling Materials (balance):

- Aluminum oxide

- Siliceous earth (diatomite)

- Color mixing materials

After paper formation, the material is impregnated with resin, with glue content of 23-30% .

2.2 Alternative Friction Material Compositions

Rubber-Based Wet Friction Materials:

For high-elasticity applications, rubber-based wet friction plates may comprise: isoprene rubber, silicone rubber, or fluorine rubber (40-50%); polystyrene or polyvinyl chloride (5-10%); sulfur, sulfide, or metal oxides (10-20%); carbon black (8-15%); thiazole accelerant (0.5-1%); anti-aging agent (0.5-1%); and functional fillers (balance) .

Ceramic-Metallic (Cera-Metallic) Materials:

For heavy-duty agricultural applications, sintered cera-metallic pads may contain: copper powder (15-25%), iron powder (25-35%), tin powder (2-3%), solid lubricants (12-15%), thermally stable inert fillers (1-2%), and inorganic abrasives (5-8%) .

Resin Mold Type Materials:

These materials use thermosetting resins (5-40% by weight) as binder, with fillers including rubber and exfoliated graphite. Phenol resin, epoxy resin, urea resin, melamine resin, and polyimide resin are commonly used .

2.3 Water-Laid Friction Material Formulations

For asbestos-free water-laid friction facing materials suitable for tractor applications, the preferred composition by weight is :

- Wood pulp: 3-7%

- Diatomite: 12-25%

- Cardolite (cashew nut shell liquid reaction product): 6-18%

- Inorganic silicate fibers (mineral wool or alumino-silicate ceramic fibers): 25-50%

- Thermosetting phenolic resin: 20-40%

- Graphite (optional): 10-30%

Example formulation for agricultural tractor brakes :

- Wood pulp: 4.2%

- Mineral wool: 24.8%

- Diatomite: 16.8%

- Cardolite: 9.4%

- Resin: 24.8%

- Graphite: 20.0%

3. Equipment

3.1 Mixing and Blending Equipment

- **V-Blender**: For preparing homogeneous powder mixtures

- **Cone blender**: For blending metallic and non-metallic powders (e.g., 30-minute blending time)

- **Kneader or Banbury mixer**: For rubber-based formulations

- **Hollander beater (Valley beater)**: For paper-based material preparation, operating at 2000-2800 rpm

3.2 Forming Equipment

- **Cold press**: For compacting powders into briquettes (pressure: 2-3 tonne/cm² or 200-600 MPa)

- **Hydraulic press**: For hot pressing operations

- **Hand sheet machine**: For paper sheet formation (diameter typically 400mm)

- **Mold cavity and die**: Shaped to desired friction plate contour

3.3 Sintering and Curing Equipment

- **Muffle furnace**: For sintering operations (temperatures up to 800°C)

- **Vulcanizer**: For curing rubber-based friction plates (temperature: 180-185°C)

- **Hot press**: For simultaneous heat and pressure application (may be water-cooled to maintain exact curvature)

3.4 Finishing Equipment

- **Punch press**: For steel plate preparation (inner/outer circle punching, hole punching)

- **Grinding machine**: For surface grinding and chamfering

- **Spreader**: For applying adhesive to central layer

4. Manufacturing Processes

4.1 Paper-Based Wet Friction Plate Process

The manufacturing process for paper-based wet brake friction plates involves the following sequential steps :

Step 1: Batching and Pulping

Water and ingredients are combined at a ratio of 200:8. In a Hollander beater, water is injected first, then cotton pulp is added. After the cotton pulp is broken down, carbon fiber and synthetic fibers are added. Other ingredients including graphite, sepiolite, friction powder, iron oxide red, iron oxide yellow, and feldspar are then beaten carefully until the slurry is refined. Latex and sulfur vulcanization agents are added, followed by sheet alkali for solidification. The beating process requires approximately 3 hours.

Step 2: Paper Formation (Sheet Making)

The beaten mixture is transferred to a hand sheet machine. The wet lapped paper thickness is set to 60% of the finished product thickness. After pouring the weighed slurry, a vacuum suction pump drains the water, forming the paper sheet.

Step 3: Drying

The paper sheets are placed in a dryer at 100°C with continuous turning during the drying process.

Step 4: Impregnation

Paper scraps are packed in a basket with chain-link fence separators. The basket is submerged in a size applicator containing a mixture of 30% cashew nut oil and 70% anhydrous alcohol. Vacuum is applied at 0.07-0.08 Pa for 15-20 minutes, then glue is added. Impregnation continues for 1.5-2 hours until complete penetration is achieved.

Step 5: Drying and Solidification

The impregnated paper is dried again and then solidified.

Step 6: Central Layer Preparation

A steel plate is punched with inner and outer circles, and 6 holes of 30mm diameter are punched. The plate is heat-treated to achieve HRC 48-53 hardness. Both sides are ground, cleaned, and degreased with acetone. Adhesive is applied and dried.

Step 7: Pasting and Vulcanization

Paper scraps are joined to both sides of the central layer. The assembly is placed on a vulcanizer flat board with height limit pieces at the four corners to ensure parallelism. Vulcanization conditions: temperature 180-185°C, pressure 38-40 MPa, dwell time 7-8 minutes.

Step 8: Post-Processing

The solidified friction plate undergoes chamfering, flat surface grinding, and oil groove machining.

Step 9: Cleaning and Packing

Residual glue marks and floating dust are removed, then the product is tested and packed.

4.2 Sintered Cera-Metallic Pad Process

For heavy-duty tractor applications using cera-metallic materials, the powder metallurgy route is employed :

Step 1: Formulation and Weighing

Raw materials are weighed according to the required composition.

Step 2: Blending

A V-blender prepares a homogeneous mix of the powders.

Step 3: Cold Pressing

The homogeneous mix is subjected to cold pressing using an appropriate mold and press at 2-3 tonne/cm² pressure to form briquettes.

Step 4: Back Plate Preparation

Copper back plate is prepared for the sintering process.

Step 5: Sintering

The briquette is compressed through the copper back plate at 800°C for 1-1.5 hours at approximately 2 MPa pressure.

Step 6: Finishing

Drilling of holes and final machining operations.

4.3 Alternative Processing: Cold Pressing, Sintering, and Hot Pressing

For certain friction material formulations, a three-stage process may be used :

Cold Pressing: Room temperature, pressure 6.75-7.0 MPa, duration 20 minutes.

Sintering: Temperature 185-190°C, atmospheric pressure, duration 20 minutes.

Hot Pressing: Temperature 205°C, pressure 6.75-7.0 MPa, duration 40 minutes.

5. Key Process Parameters and Quality Considerations

Porosity Control: Suitable porosity of friction material is preferably in the range of 10-40%. When porosity is 50% or more, strength deteriorates and the friction layer is liable to break under shear forces . Pore diameter should be sufficiently large to prevent clogging during sliding, with larger pores at the outer surface and gradually smaller pores nearer the core plate .

Bonding: The friction material must be strongly bonded to the steel backing plate. The glue composition (30% modified resin and 70% anhydrous alcohol) and vulcanization parameters (180-185°C, 38-40 MPa, 7-8 minutes) are critical for achieving proper adhesion .

Thermal Stability: For heavy-duty wet friction materials, material strength at 200-250°C is very important. Thermosetting resins such as phenol, epoxy, and polyimide resins provide the necessary mechanical integrity under these temperature conditions .

6. Summary Table

Component	Materials	Equipment	Process Conditions
Fiber Reinforcement	Aramid, carbon, cotton pulp	Hollander beater	2000-2800 rpm, 3 hours
Binder	Phenolic resin, rubber	Impregnation tank	Vacuum 0.07-0.08 Pa, 1.5-2 hours
Friction Modifiers	Diatomite, graphite, cardolite	V-blender	30 minutes blending
Functional Fillers	Al_2O_3, iron oxides, feldspar	Hand sheet machine	400mm diameter
Backing Plate	Steel or copper	Punch press	HRC 48-53
Curing	—	Vulcanizer	180-185°C, 38-40 MPa, 7-8 min
Sintering	—	Muffle furnace	800°C, 1-1.5 hours